

UNIVERSITÀ
DEGLI STUDI
DELL'AQUILA

Lesson 8: Unsupervised Learning

Technologies for Artificial Intelligence

Fabio Antonini, Università degli Studi dell'Aquila

27 November 2026

Agenda

- No target at all: what changes
- k-means: one centre per group, two moves, how many groups
- When groups are not round: a tree of merges, and DBSCAN
- Principal component analysis (PCA): the widest shadow
- Fraud that looks ordinary on every column, and how to catch it

UNIVERSITÀ DEGLI STUDI DELL'AQUILA

Exercise 7, before we start

- Nothing is collected: we discuss it, ten minutes
- What counts is the **defence of the chosen model family**, not the winning score
- The common gap: an importance ranking quoted as fact, without checking how much of it landed on noise columns
- Today the same duty, with no labels at all: every grouping is a claim to check

No target at all

- So far: pairs, an input x and a target y
- **Unsupervised learning** has only x: no answer key
- The task: “describe the structure”, not “predict the label”
- Most data was never labelled: too slow, costly, or ill defined
- So these methods often run **first**

UNIVERSITÀ DEGLI STUDI DELL'AQUILA

Aurora: three problems, no answer key

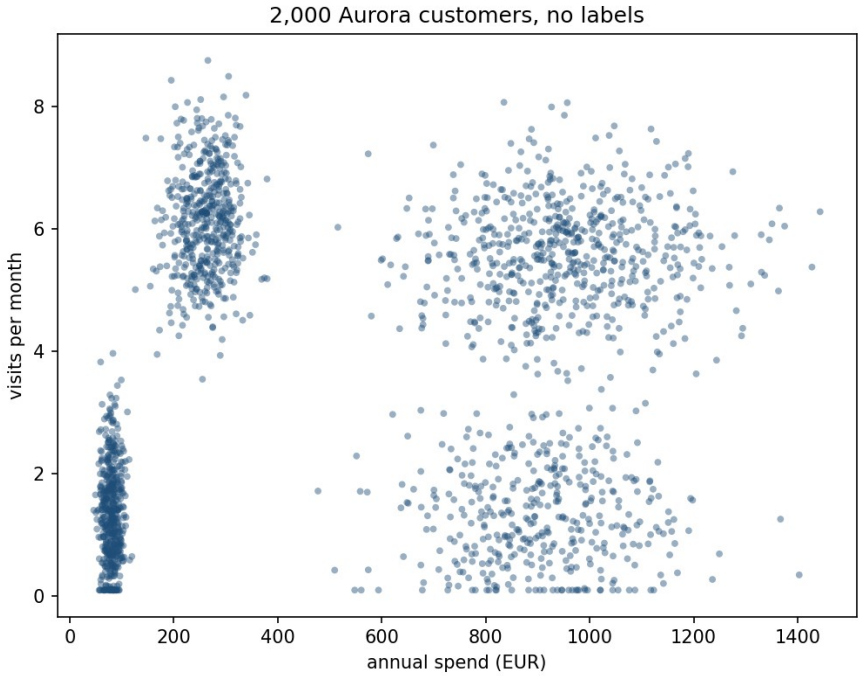
- **Group 2,000 customers**, from annual spend and visits per month: **k-means**, notebook 1
- **Tell bots from humans** in 1,500 website sessions with the same average behaviour: **a tree of merges and DBSCAN**, notebook 2
- **Catch fraud** in an 8-column account table, where every value looks ordinary: **PCA**, notebook 3

A truth Aurora will never have

- `Notebooks/retail_data.py` makes all three datasets, and publishes how in `TRUE_*` constants
- So today every method can be checked against the real answer
- Aurora’s own analysts could never do that check
- Remember the difference: it makes the lesson teachable, and it is exactly what is missing on real data

UNIVERSITÀ DEGLI STUDI DELL'AQUILA

2,000 customers, no labels



UNIVERSITÀ DEGLI STUDI DELL'AQUILA

One centre per group

- One point per group, its **centre**: the “typical customer”
- Every customer goes to the **nearest** centre
- **k-means** finds k centres that keep customers close to their own
- k, the number of groups, is **your choice**
- “Nearest” is straight-line distance: **standardise** first (lesson 2)

UNIVERSITÀ DEGLI STUDI DELL'AQUILA

Two moves, repeated

- Start: k centres, e.g. k random customers
- **Assign**: every point to its nearest centre
- **Move**: every centre to the **mean** of its points
- Repeat until a round changes nothing
- In code: `KMeans(n_clusters=4)`, then `labels_`, `cluster_centers_`

UNIVERSITÀ DEGLI STUDI DELL'AQUILA

One number for how tight the groups are

- Draw a line from every point to the centre of its group
- **Within-cluster sum of squares** (WCSS): the squared lengths of those lines, added up
- Short lines, small WCSS: every point is close to its centre
- scikit-learn reports it as `inertia_`

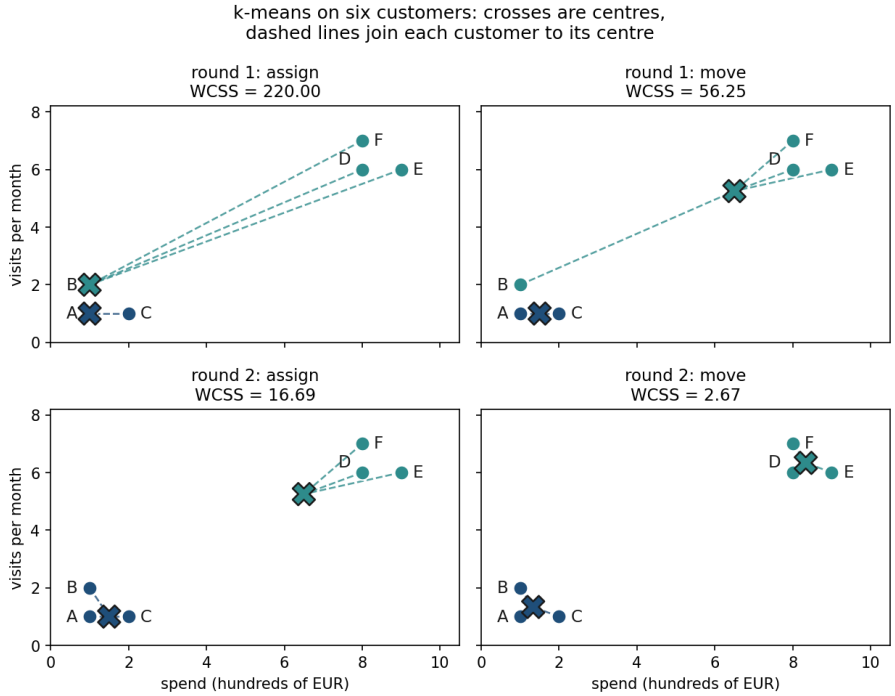
UNIVERSITÀ DEGLI STUDI DELL'AQUILA

The within-cluster sum of squares, as a formula

$$J = \sum_{i=1}^{m} \|x_i - \mu_{c(i)}\|^2$$

UNIVERSITÀ DEGLI STUDI DELL'AQUILA

Six customers, by hand

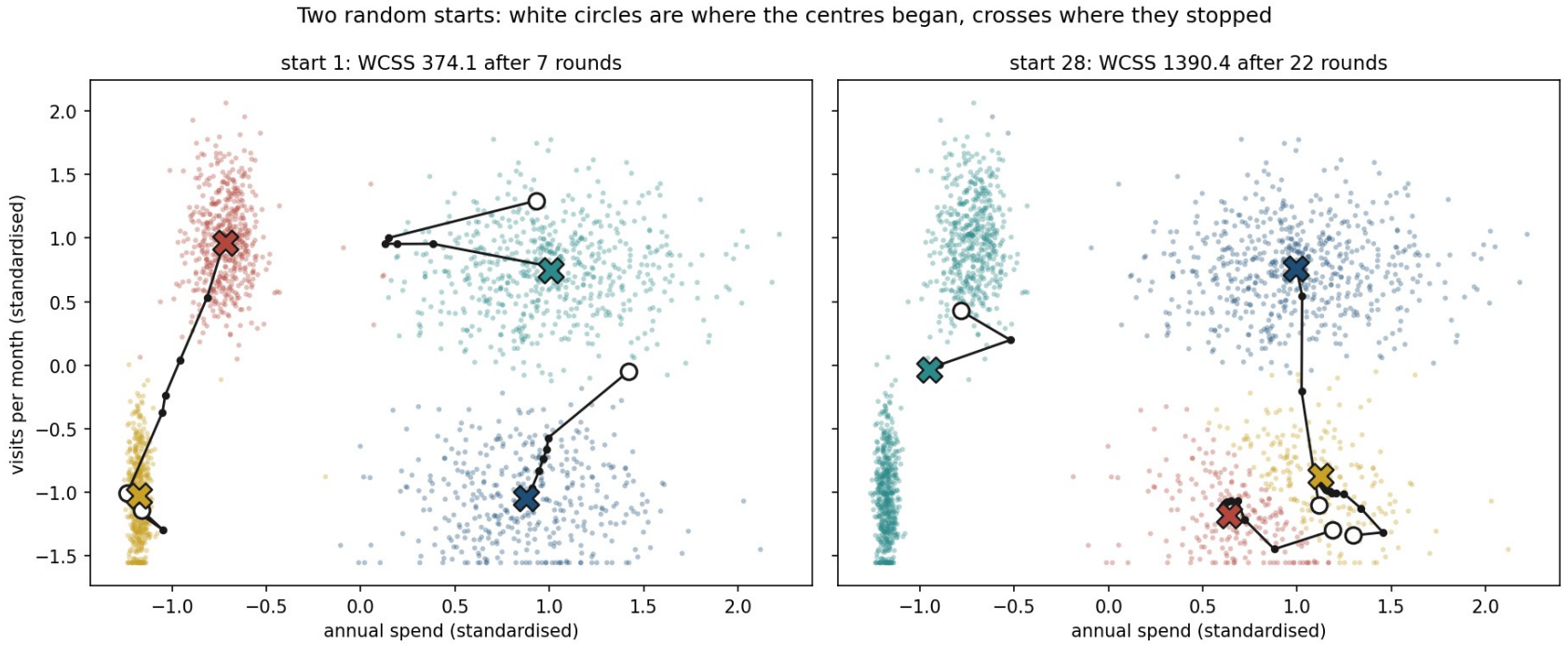


Why it always stops

- **Assign** cannot raise WCSS: points only move to a nearer centre
- **Move** cannot raise it: the mean is the closest point to a group, in squared distance
- Finitely many ways to split m points into k groups
- So WCSS falls or stays until nothing changes: it **stops**
- Six customers: 220, 56.25, 16.69, 2.67

UNIVERSITÀ DEGLI STUDI DELL'AQUILA

Same data, two starts



UNIVERSITÀ DEGLI STUDI DELL'AQUILA

The remedy: several starts, keep the best

- 30 random starts: 29 reach WCSS **374.1**, one stops at **1,390.4**
- Rare, but one run cannot tell you it was the unlucky one
- So: several starts, keep the lowest WCSS
- scikit-learn: `n_init=10`, plus smarter starts (**k-means++**)
- `KMeans` and the ten lines agree: 374.0919

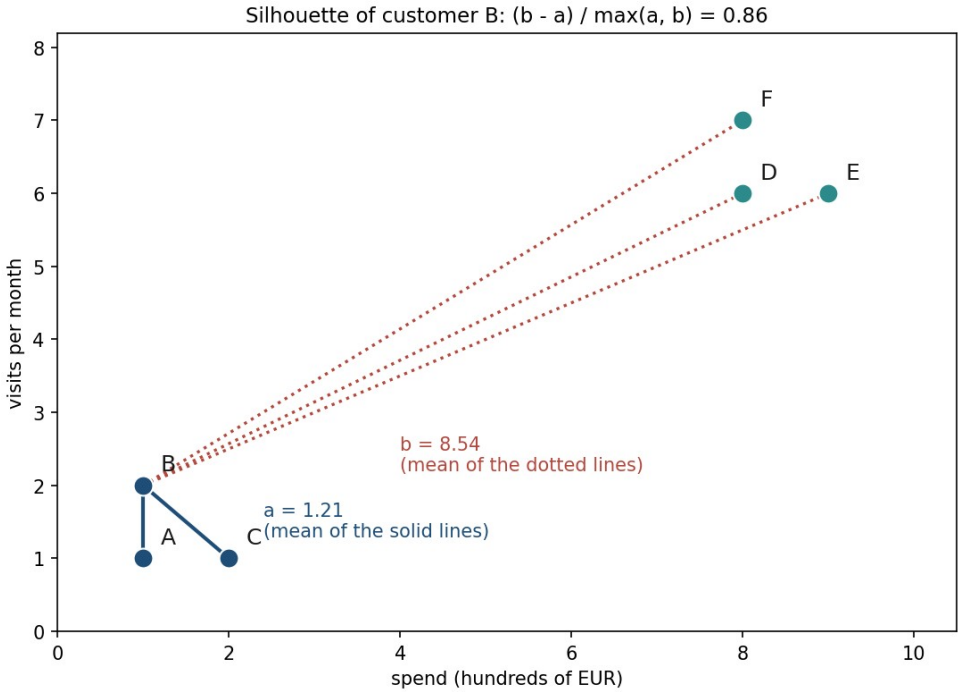
UNIVERSITÀ DEGLI STUDI DELL'AQUILA

How many groups?

- k is an **input**: the algorithm does not choose it
- WCSS always falls as k grows: one group per point gives 0
- So “lowest WCSS” always picks the largest k tried
- Instead, **the elbow**: where WCSS stops falling steeply
- And **the silhouette**: is each point nearer its own group than the next?

UNIVERSITÀ DEGLI STUDI DELL'AQUILA

The silhouette of one point



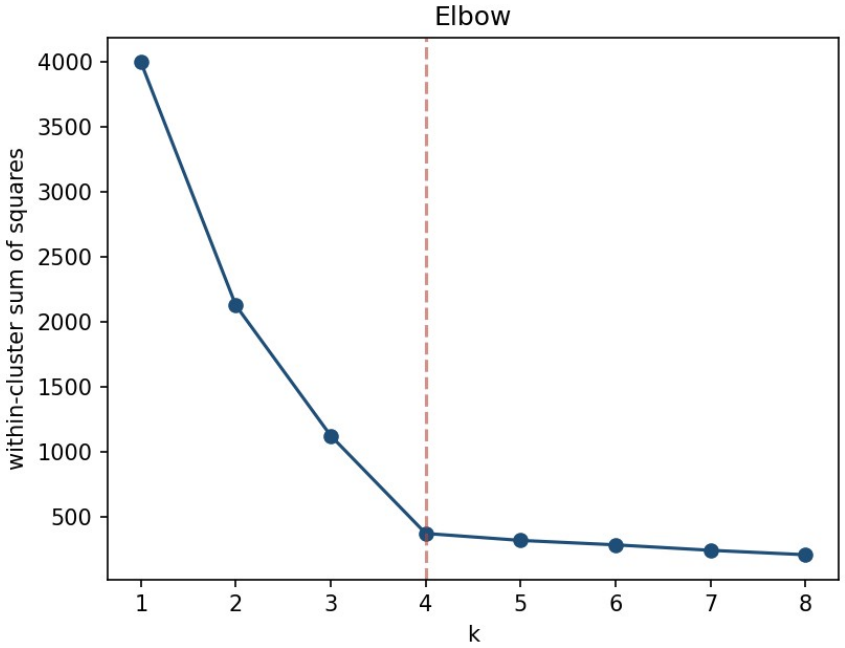
UNIVERSITÀ DEGLI STUDI DELL'AQUILA

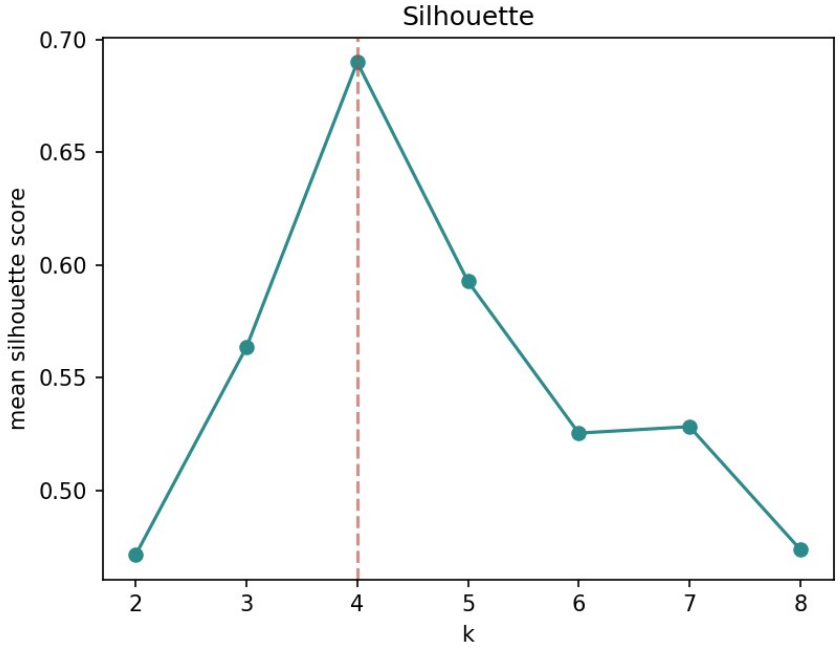
The silhouette, as a formula

$$s = \frac{b - a}{\max(a, b)}$$

UNIVERSITÀ
DEGLI STUDI
DELL'AQUILA

Elbow and silhouette, on 2,000 customers





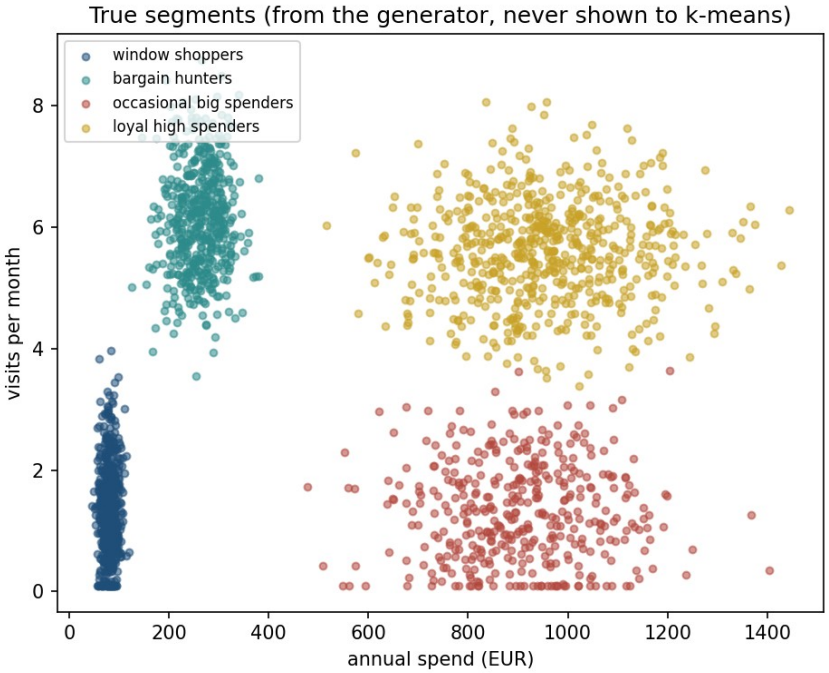
UNIVERSITÀ DEGLI STUDI DELL'AQUILA

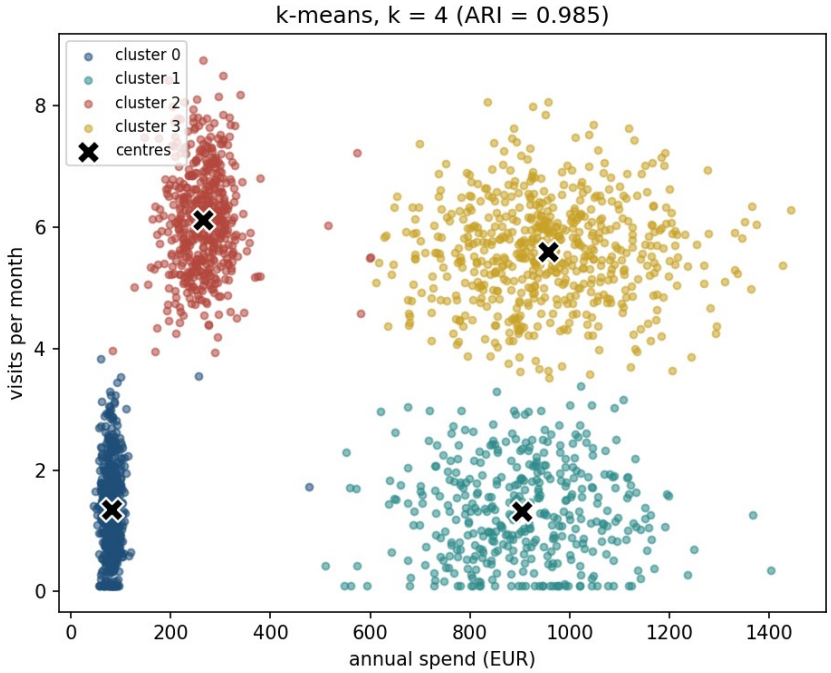
Checking against the truth: the adjusted Rand index

- Possible only here: the generator knows each customer’s segment
- For every **pair**: do both groupings agree (together, or apart)?
- Share of pairs that agree: the Rand index
- **Adjusted Rand index** (ARI): random grouping about 0, identical 1
- In code: `adjusted_rand_score(truth, labels)`

UNIVERSITÀ DEGLI STUDI DELL'AQUILA

k-means against the truth





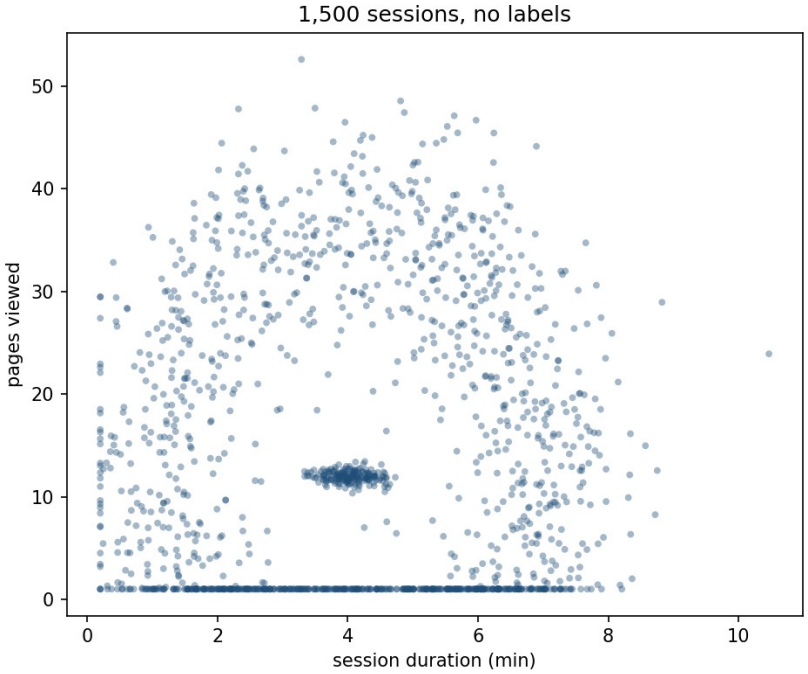
UNIVERSITÀ DEGLI STUDI DELL'AQUILA

Notebook 1, live

section, and its slides	you call (→ the number to come back with)
look first, 7	`StandardScaler` → 2,000 customers, 2 columns
k-means by hand, 9-13	NumPy only → WCSS 220, 56.25, 16.69, 2.67
two starts, 14-15	NumPy, then `KMeans` → 374.1 against 1,390.4
how many groups, 16-19	`KMeans`, `silhouette_score` → 0.690 at k = 4
against the truth, 20-21	`adjusted_rand_score` → 0.985

UNIVERSITÀ DEGLI STUDI DELL'AQUILA

1,500 sessions, no labels



UNIVERSITÀ
DEGLI STUDI
DELL'AQUILA

Why k-means must fail here

- With k = 2, the points equally far from the two centres form a **straight line**
- So k-means' two groups are always the two sides of a straight line
- No straight line separates a core from the ring around it
- Result: ARI **−0.046**, no better than chance
- Not bad luck with the start: more starts cannot fix it

UNIVERSITÀ DEGLI STUDI DELL'AQUILA

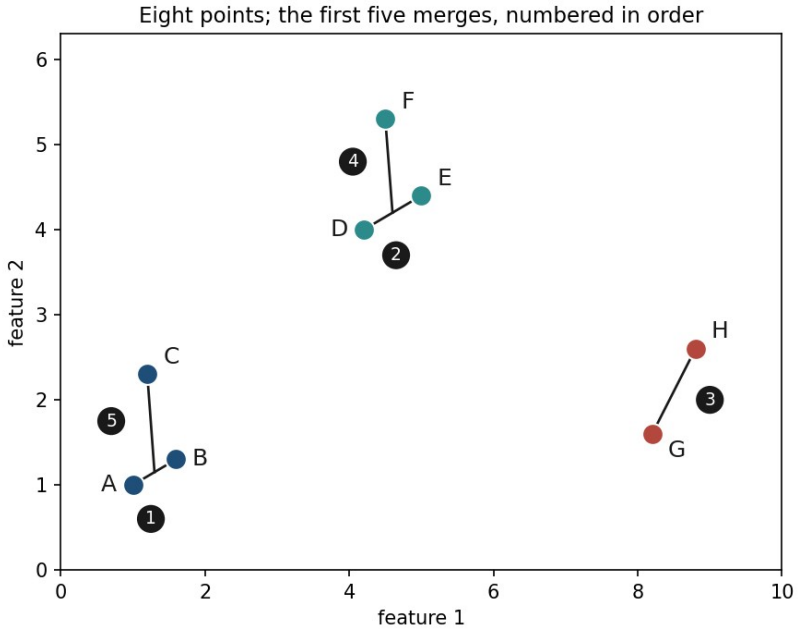
Hierarchical clustering: a tree of merges

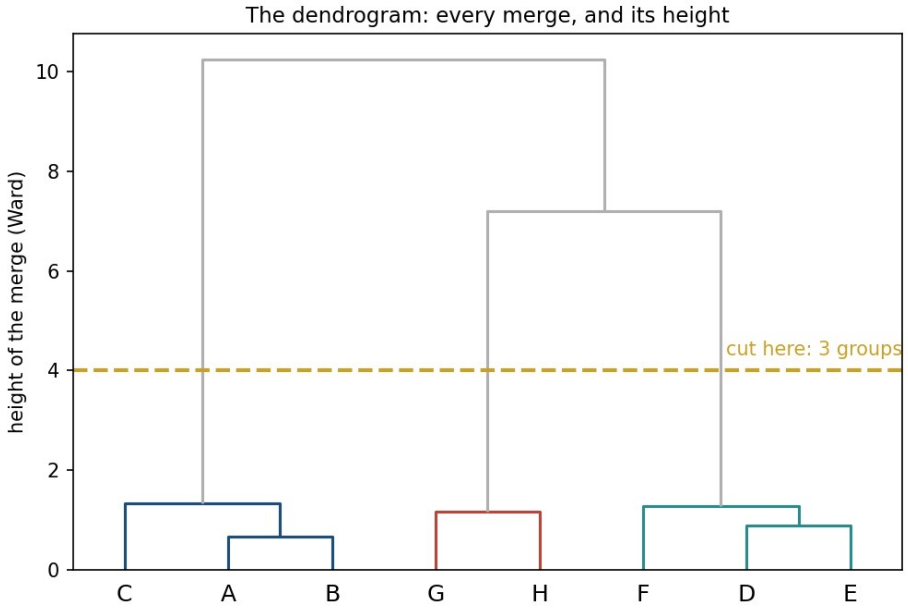
- Start: every point is a group of its own
- Merge the two **closest** groups; repeat until one is left
- The record is a tree, a **dendrogram**: a join's height = how far apart
- **Cut** it with a horizontal line: the cut chooses k
- In code:

```
AgglomerativeClustering(n_clusters=2)
```

UNIVERSITÀ DEGLI STUDI DELL'AQUILA

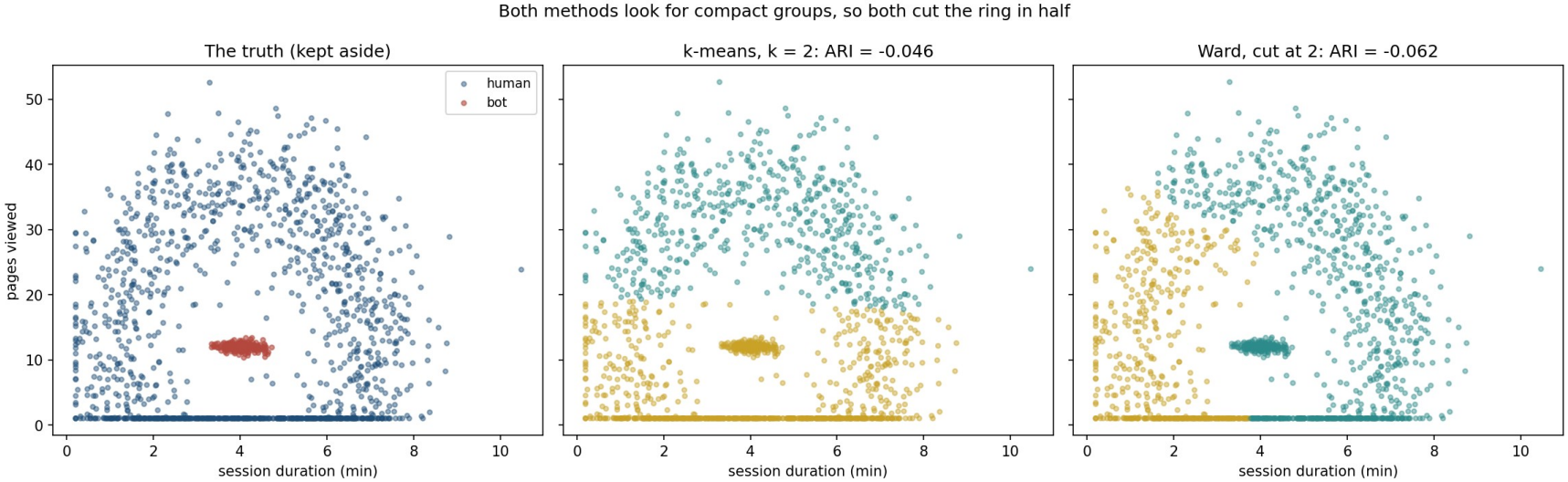
Eight points, five merges, one cut





UNIVERSITÀ DEGLI STUDI DELL'AQUILA

Ward on the sessions: the same failure



UNIVERSITÀ DEGLI STUDI DELL'AQUILA

Break

- Twelve minutes

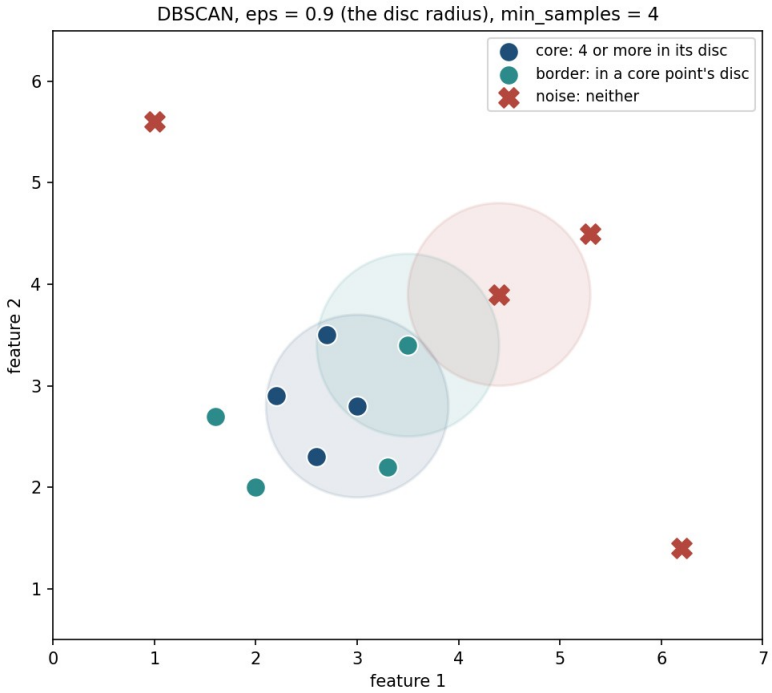
UNIVERSITÀ
DEGLI STUDI
DELL'AQUILA

DBSCAN: is this point in a crowded place?

- Look at the disc of radius `eps` around each point
- **Core** point: at least `min_samples` points in its disc, **itself included**
- **Border** point: not core, but inside a core point’s disc; it joins that group
- **Noise**: neither; it belongs to no group (label -1)
- A group is a chain of core points plus their border points: **no centres, no k**

UNIVERSITÀ DEGLI STUDI DELL'AQUILA

Core, border and noise, drawn



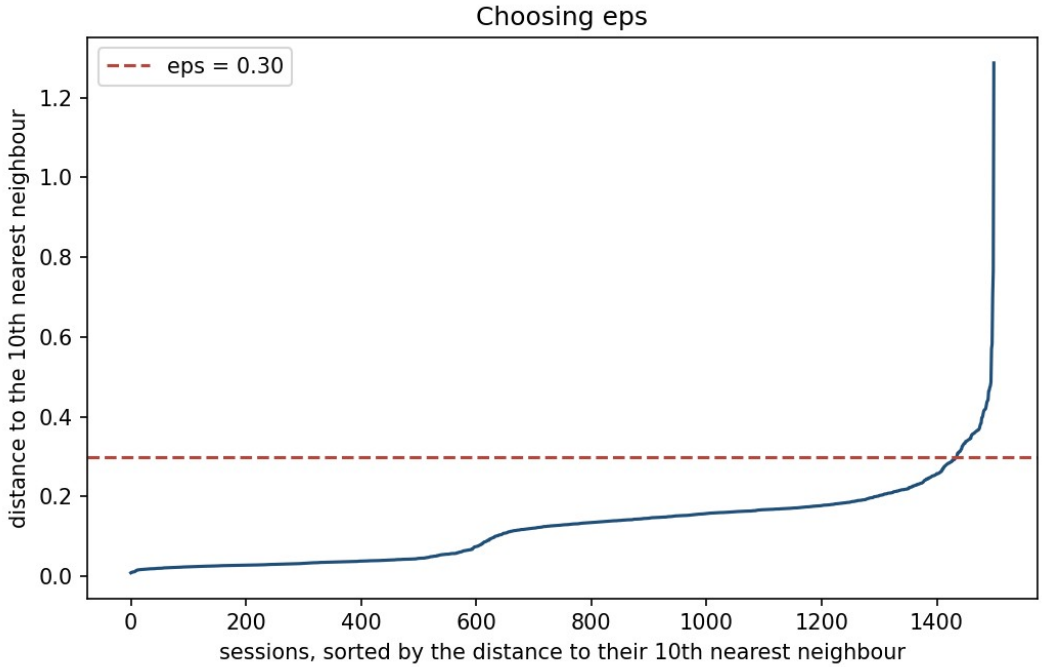
UNIVERSITÀ DEGLI STUDI DELL'AQUILA

Choosing `eps`: the k-distance plot

- Too small: even the bot core breaks into noise
- Too large: the ring reaches the core, all one group
- Sort every point’s distance to its 10th neighbour (10 = `min_samples`)
- Put `eps` where the sorted curve **bends upward**
- In code: `NearestNeighbors(n_neighbors=10)`, then `kneighbors`

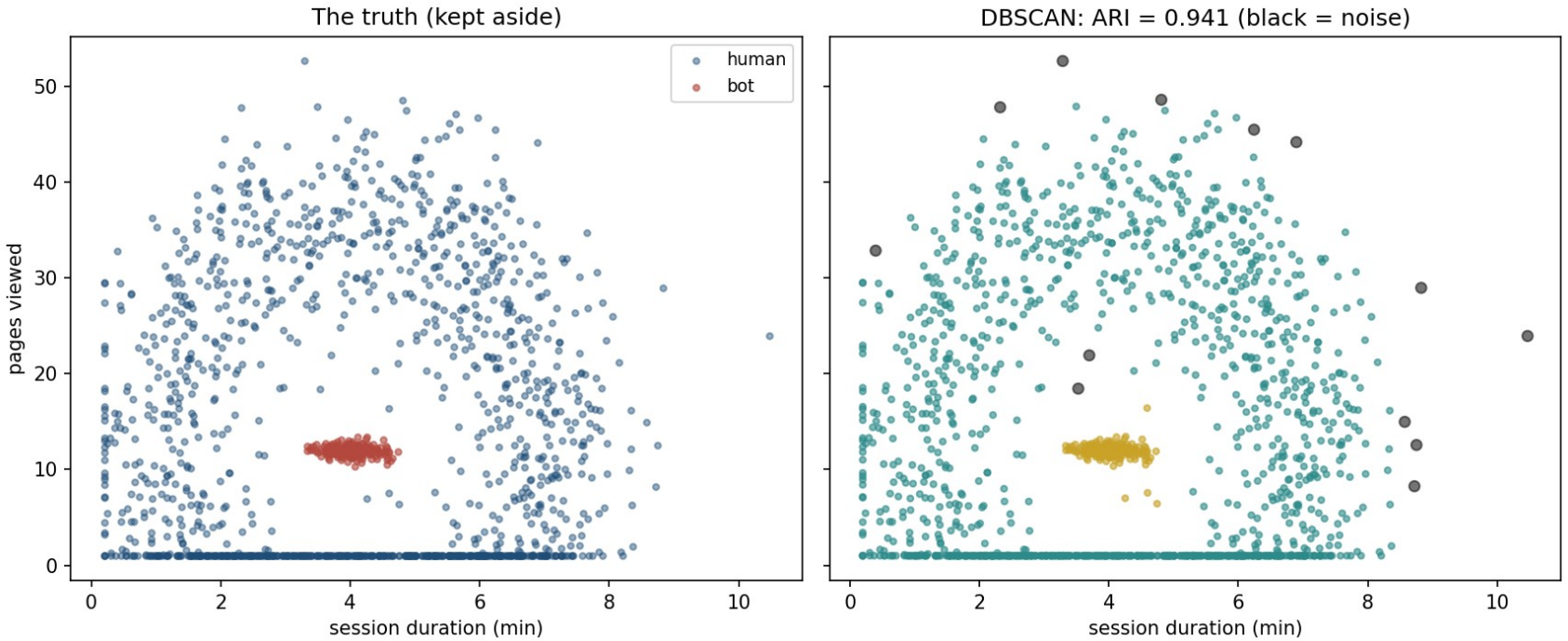
UNIVERSITÀ DEGLI STUDI DELL'AQUILA

The bend in the curve



UNIVERSITÀ DEGLI STUDI DELL'AQUILA

DBSCAN on the sessions



UNIVERSITÀ DEGLI STUDI DELL'AQUILA

The 13 noise sessions

- All 13 are **humans**
- They sit where the data is thin: 2 in the gap around the bot core, 11 on the ring’s sparse outer edge
- DBSCAN’s answer for them: **not confidently either**
- For a security analyst, more useful than a forced “bot” or “human”

The silhouette prefers the wrong answer

grouping	silhouette (no truth needed)	ARI (needs the truth)
k-means, k = 2	0.401	−0.046
DBSCAN	−0.106	0.941

UNIVERSITÀ DEGLI STUDI DELL'AQUILA

Choosing among the three

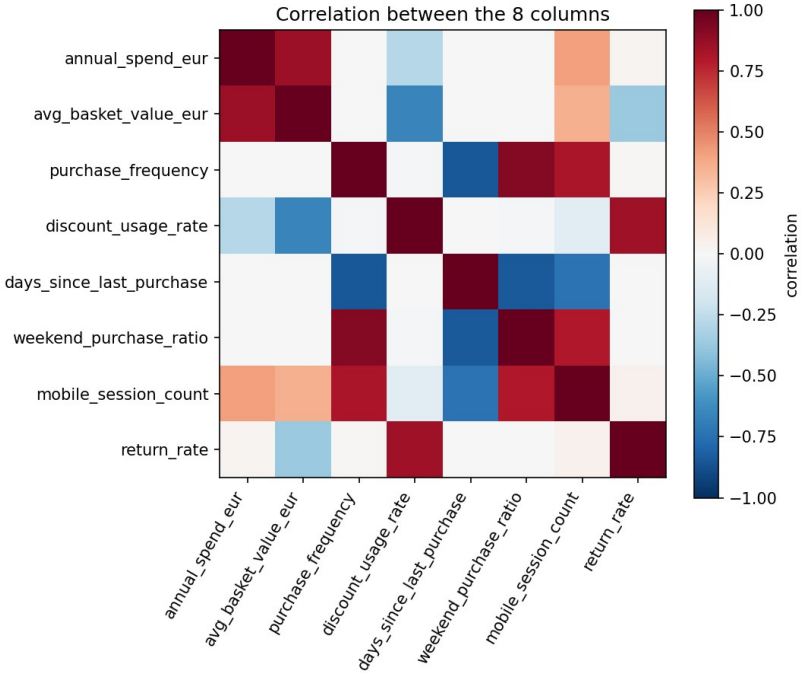
method	a group is	needs k?	fails when
k-means	the points nearest one centre	yes	groups are not round
tree of merges, Ward	a branch of the tree	the cut chooses it	as k-means
DBSCAN	a crowded connected region	no	groups differ a lot in density

UNIVERSITÀ DEGLI STUDI DELL'AQUILA

Notebook 2, live

section, and its slides	you call (→ the number to come back with)
look first, 23	`StandardScaler` → 1,500 sessions, 180 bots
k-means, 24	`KMeans`, `adjusted_rand_score` → ARI −0.046
tree of merges, 25-27	`AgglomerativeClustering` → ARI −0.062
DBSCAN drawn, 29-30	`DBSCAN` → 4 core, 4 border, 4 noise
`eps`, 31-32	`NearestNeighbors` → 65 sessions beyond 0.30
DBSCAN, 33-34	`DBSCAN` → ARI 0.941, 13 noise
silhouette, 35	`silhouette_score`, `silhouette_samples` → 0.401 against −0.106

Eight columns, not eight independent numbers



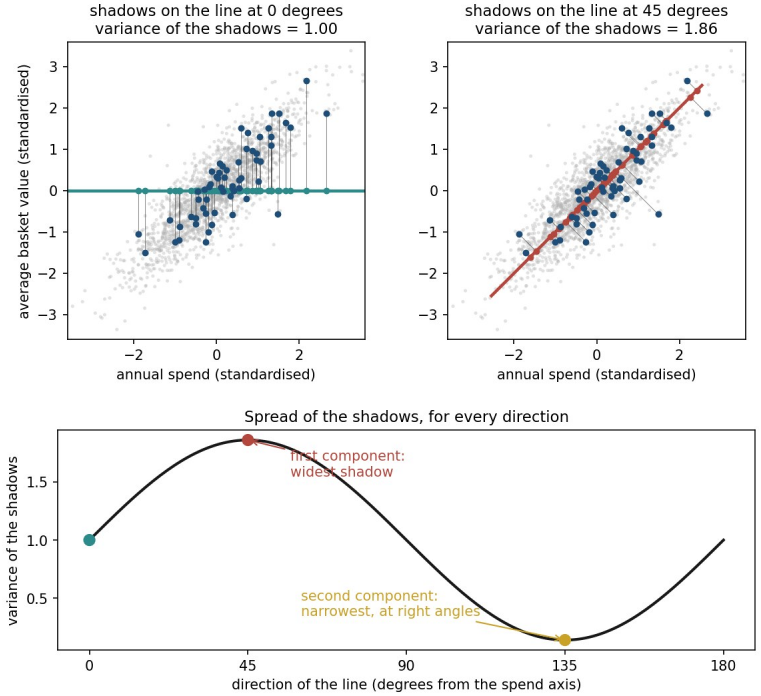
UNIVERSITÀ DEGLI STUDI DELL'AQUILA

The direction with the widest shadow

- Spend and basket value, standardised: correlation 0.859
- Drop each point onto a line at a right angle: its **shadow**
- One number per account sums up two columns
- Best line: shadows **spread most** (largest variance)
- That is the **first principal component**

UNIVERSITÀ DEGLI STUDI DELL'AQUILA

The spread of the shadows



UNIVERSITÀ
DEGLI STUDI
DELL'AQUILA

Why the peak is 1 + r

$$\mathrm{Var}(z_\theta) = 1 + r\sin 2\theta$$

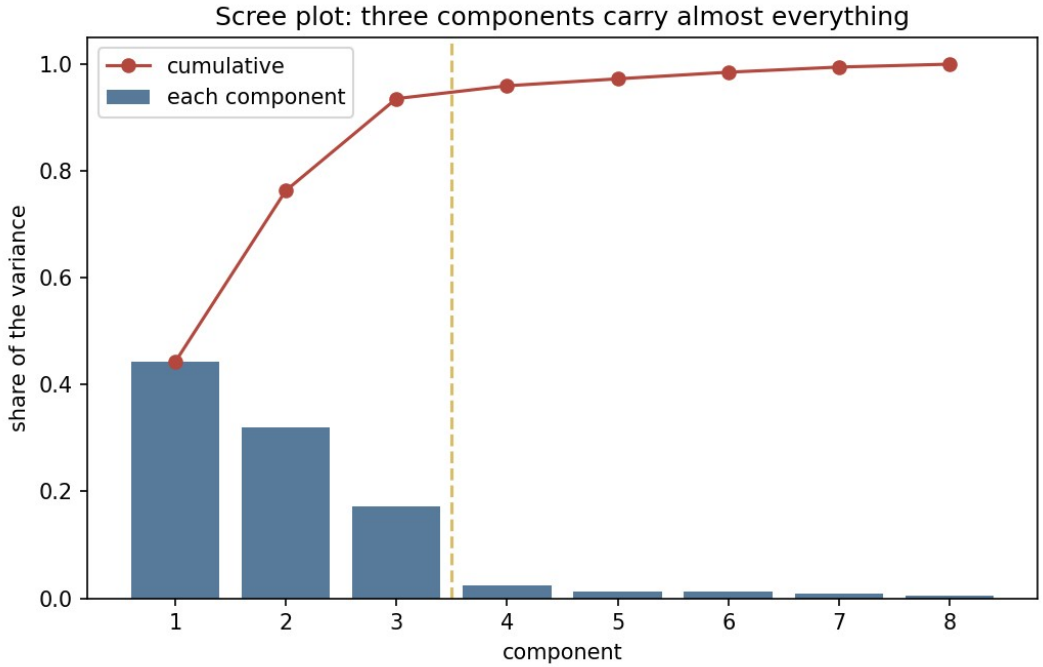
UNIVERSITÀ
DEGLI STUDI
DELL'AQUILA

Any number of columns: stated, not derived

- 8 columns: no angle to sweep
- **The components are the eigenvectors of the covariance matrix**
- Each **eigenvalue**: the variance of the shadows along it
- Two columns: 1 + r and 1 - r, on the diagonals, as swept
- In code: `PCA().fit(X)`, shares in `explained_variance_ratio_`

UNIVERSITÀ DEGLI STUDI DELL'AQUILA

How many components: the scree plot



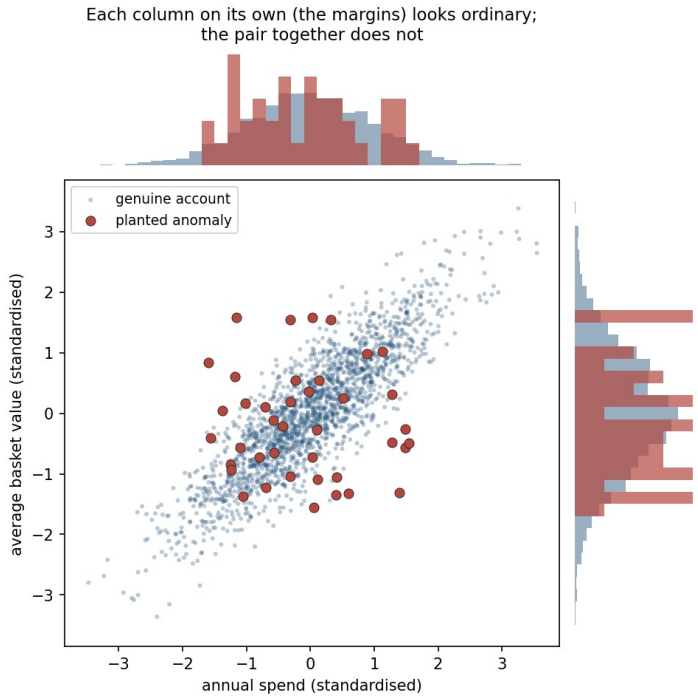
UNIVERSITÀ DEGLI STUDI DELL'AQUILA

Fraud that looks ordinary on every column

- 40 of 2,000 accounts (2%) taken over by fraudsters
- Each value drawn **inside** the genuine 5th-95th percentile range, column by column: notebook 3 finds 100% inside
- So no one-column rule (“spend above X”) finds them
- What fraud cannot copy: **how the columns move together**

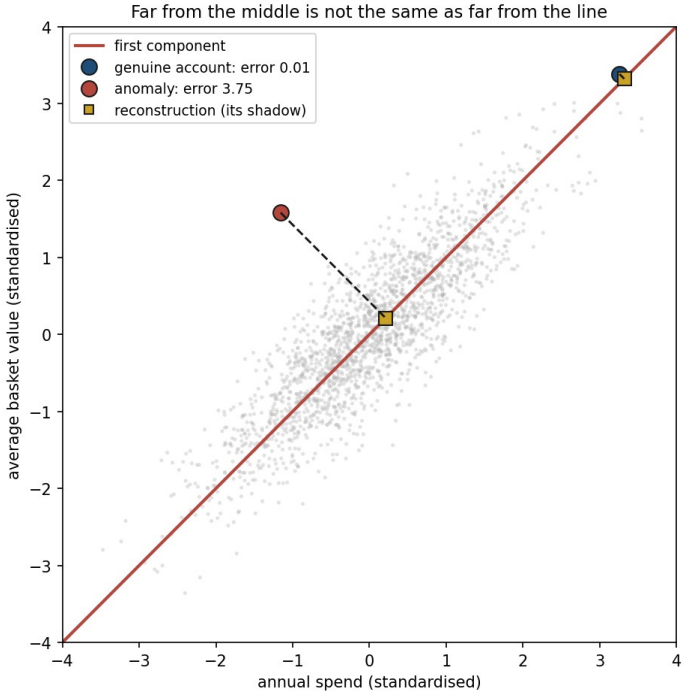
UNIVERSITÀ DEGLI STUDI DELL'AQUILA

Off the band



UNIVERSITÀ
DEGLI STUDI
DELL'AQUILA

Rebuild each account from its shadow



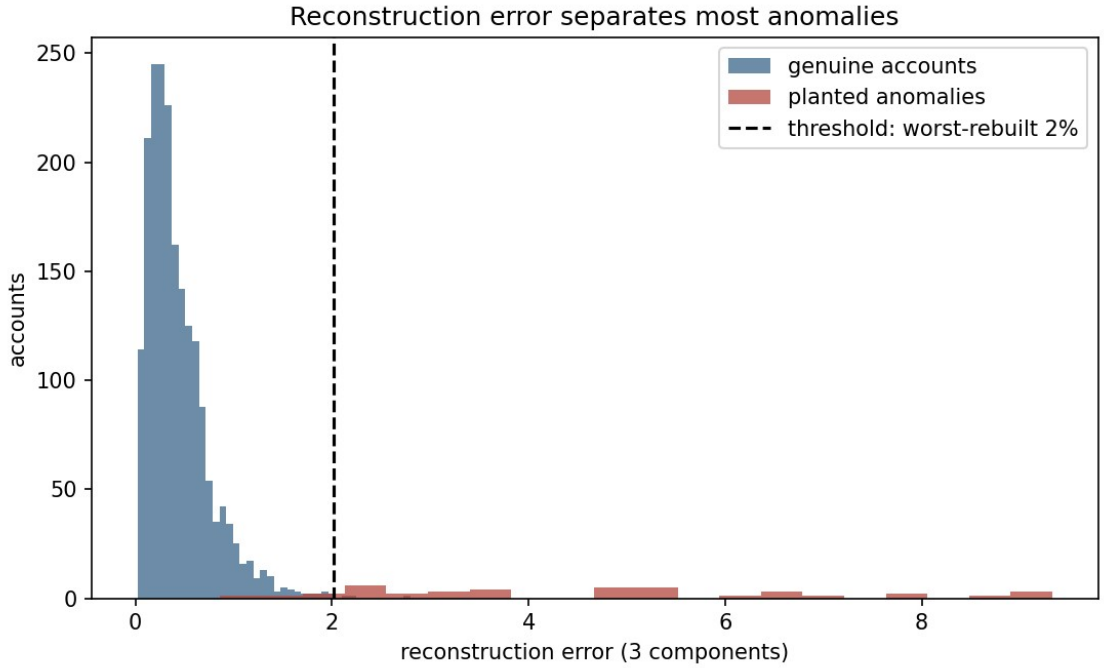
UNIVERSITÀ
DEGLI STUDI
DELL'AQUILA

The reconstruction error, as a formula

$$e = \|x - \hat{x}\|^2$$

UNIVERSITÀ
DEGLI STUDI
DELL'AQUILA

37 of 40



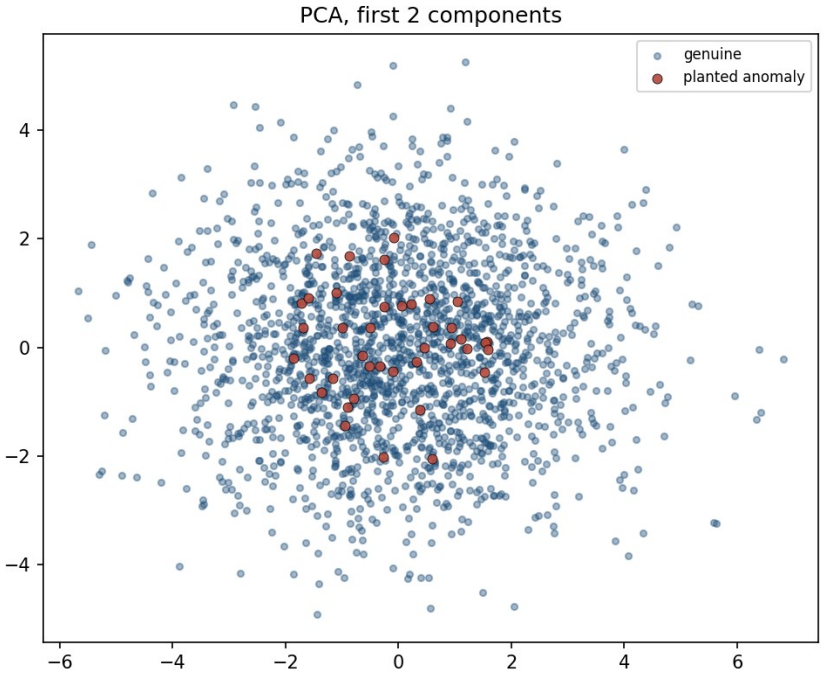
UNIVERSITÀ DEGLI STUDI DELL'AQUILA

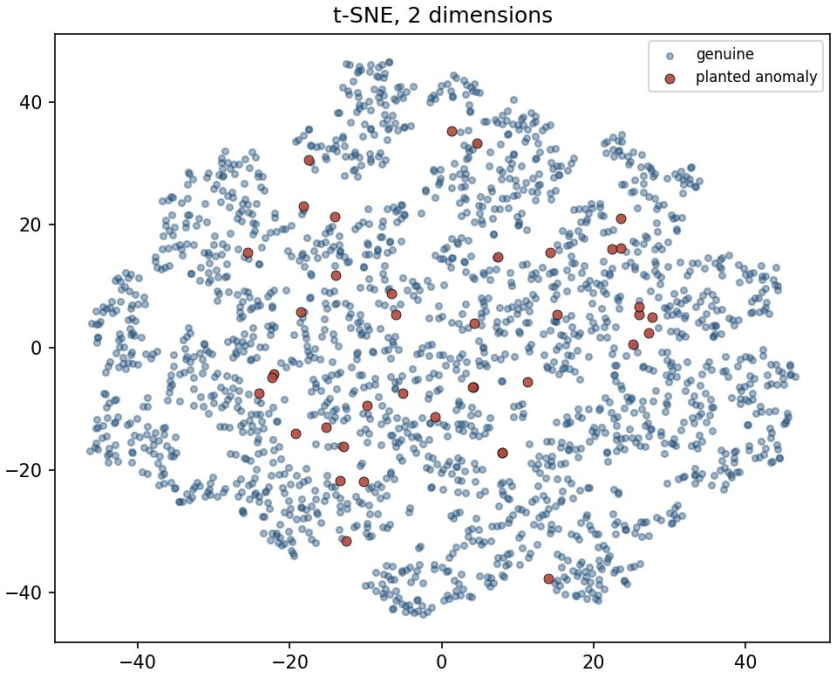
37 is one batch

- Regenerate the table at 7 other seeds: same process, different accounts
- Same procedure: 3 components, flag the worst-rebuilt 2%
- Caught: from **32 to 38** of 40, median 36
- Carry 37 as the memorable figure, and the range as what it means: about nine in ten
- Lesson 5’s point, with no y: one run is one draw

UNIVERSITÀ DEGLI STUDI DELL’AQUILA

Why not just plot it and look?





UNIVERSITÀ DEGLI STUDI DELL'AQUILA

t-SNE: for looking, not for measuring

- **t-SNE** (t-distributed stochastic neighbour embedding): made for pictures
- In 2-D, **close neighbours stay close**
- Large distances are not kept: gaps **mean little**
- No `transform`: a new account means running it all again
- In code: `TSNE(n_components=2)`

UNIVERSITÀ DEGLI STUDI DELL'AQUILA

Notebook 3, live

section, and its slides	you call (→ the number to come back with)
eight columns, 38	`StandardScaler` → correlation 0.859
the widest shadow, 39-42	NumPy, then `PCA` → 1.859 and 0.141; shares 0.9297 and 0.0703
scree, 43	`PCA().fit`, `explained_variance_ratio_` → 93.5% in 3 components
fraud, 44-47	`PCA(n_components=1)`, `inverse_transform` → errors 0.01 and 3.75
37 of 40, 48-49	`PCA(n_components=3)` → 37 of 40; 32 to 38 over 8 batches
2-D pictures, 50-51	`PCA(n_components=2)`, `TSNE` → 1.30 against 2.20

UNIVERSITÀ DEGLI STUDI DELL'AQUILA

What to take away

- k-means always stops, not always at the best: **374.1** against **1,390.4**
- A core inside a ring defeats k-means and Ward; DBSCAN scores **0.941**
- The silhouette measures compactness: it preferred the wrong answer
- **37 of 40** disguised accounts caught by reconstruction error, invisible to a 2-D scatter

Homework

- **Exercise 8**, discussed at the start of lesson 9, **Friday 4 December**
- `Exercises/08_unsupervised_learning.md`
- Two account books: predict, **before plotting**, where a 2-D plot will find the fraud and where it will not

UNIVERSITÀ DEGLI STUDI DELL'AQUILA